

Light, Frequency and the Photoelectric Effect

PhET Photoelectric Effect · PhET

Senior Secondary (Years 11-12)

~30 minutes

Science · Physics

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://phet.colorado.edu/en/simulations/photoelectric>

Curriculum links: quantum physics · nuclear physics

Learning intentions

- I can describe how varying light frequency and intensity independently affects electron ejection from a metal surface.
- I can explain why classical wave theory fails to predict the photoelectric effect, while a photon (particle) model succeeds.
- I can identify the threshold frequency for a given metal and relate it to its work function.

Getting started

Open PhET Photoelectric Effect. Select a metal, starting with sodium, and set the light intensity to a low but non-zero value. You will vary frequency and intensity independently across the tasks, so keep the current meter and voltage panel visible.

Tasks

1. Start with a low light frequency, at the red end of the spectrum, at moderate intensity. Does the simulation show any electrons being ejected? Record what you observe.

2. Gradually increase the frequency, moving towards violet/UV, keeping intensity constant. At what point do electrons begin to be ejected?

3. Now return to a frequency just below this threshold and increase the intensity as high as it will go. Do electrons get ejected? Explain what this tells you about the roles of frequency versus intensity.

4. Test frequencies below, at and above the threshold, at two different intensities each, and record whether electrons are ejected and the current reading.

Frequency (below/at/above threshold)	Intensity	Electrons ejected? (Y/N)	Current reading

5. Using the simulation's energy-per-photon display, or the relationship $E = hf$, estimate the work function of your chosen metal from the threshold frequency you found.

Working:

6. Change the metal to one with a higher work function. Does the threshold frequency increase or decrease? Explain why this makes sense.

7. Explain why the photoelectric effect could not be explained by treating light purely as a wave, and how thinking of light as photons resolves the problem.

Extension (optional)

Investigate how increasing the intensity above the threshold frequency affects the current versus the maximum kinetic energy of ejected electrons. Explain why these two quantities respond differently to intensity.
